

Coal Mine Safety and Health Act 1999
Electrical Engineering Manager Legislation Exam – UG

12 Meaning of *competence*

Competence for a task at a coal mine is the demonstrated skill and knowledge required to carry out the task to a standard necessary for the safety and health of persons.

15 Meaning of *accident*

An *accident* at a coal mine is an event, or a series of events, at the coal mine causing injury to a person.

16 Meaning of *serious accident*

A *serious accident* at a coal mine is an accident at a coal mine that causes—

- (a) the death of a person; or
- (b) a person to be admitted to a hospital as an in-patient for treatment for the injury.

17 Meaning of *high potential incident*

A *high potential incident* at a coal mine is an event, or a series of events, that causes or has the potential to cause a significant adverse effect on the safety or health of a person.

26 Meaning of *supervisor*

A *supervisor* at a coal mine is a person appointed under [section 56](#) to—

- (a) implement and monitor the coal mine's safety and health management system; and
- (b) give directions to other coal mine workers at the coal mine in accordance with the safety and health management system.

30 How is an acceptable level of risk achieved

(1) To achieve an acceptable level of risk, this Act requires that management and operating systems must be put in place for each coal mine.

(2) This Act provides that the systems must incorporate risk management elements and practices appropriate for each coal mine to—

- (a) identify, analyse, and assess risk; and
- (b) avoid or remove unacceptable risk; and
- (c) monitor levels of risk and the adverse consequences of retained residual risk; and
- (d) provide for critical controls; and
- (e) investigate and analyse the causes of serious accidents and high potential incidents with a view to preventing their recurrence; and
- (f) review the effectiveness of risk control measures, and take appropriate corrective and preventive action; and
- (g) mitigate the potential adverse effects arising from residual risk.

(3) Also, the way an acceptable level of risk of injury or illness may be achieved may be prescribed under a regulation.

39A Additional obligations

(1) This section applies to each of the following persons—

- (a) a coal mine worker at a coal mine;
- (b) another person at a coal mine;
- (c) an ROC worker for a coal mine.

(2) The person has the following additional obligations—

- (a) to work or carry out the person's activities in a way that does not expose the person or someone else to an unacceptable level of risk;
- (b) to ensure, to the extent of the responsibilities and duties allocated to the person, that the work and activities under the person's control, supervision, or leadership is conducted in a way that does not expose the person or someone else to an unacceptable level of risk;
- (c) to the extent of the person's involvement—to participate in and conform to the risk management practices of the mine;
- (d) to comply with instructions given for safety and health of persons by the coal mine operator or site senior executive for the mine or a supervisor at the mine;
- (e) to work at or for the coal mine only if the person is in a fit condition to carry out the work without affecting the safety and health of others;
- (f) not to do anything wilfully or recklessly that might adversely affect the safety and health of someone else at the mine.

55 Management structure for safe operations at coal mines

(1) The site senior executive for a coal mine must—

- (a) develop and maintain a management structure for the coal mine in a way that allows development and implementation of the safety and health management system; and
- (b) document the management structure.

Maximum penalty—40 penalty units.

(2) The document must state—

- (a) the responsibilities of the site senior executive; and
- (b) the responsibilities and competencies required for senior positions in the structure; and
- (c) the names of the persons holding the senior positions and their competencies; and
- (d) the name of the person who is responsible for establishing and implementing a system for managing contractors at the coal mine; and
- (e) the competencies required, and the responsibilities, for each other supervisory position at the mine.

Maximum penalty—40 penalty units.

(3) For subsection (2)(b), an inspector may by notice given to the site senior executive declare a position to be a senior position.

(4) For each supervisory position mentioned in subsection (2)(e), the site senior executive must also keep a record of the names and competencies of each person authorised to carry out the responsibilities of the position.

60 Additional requirements for management of underground mines

(1) This section applies to an underground mine.

(2) The site senior executive must appoint a person to be underground mine manager to control and manage the mine.

Maximum penalty—400 penalty units.

(3) However, the site senior executive may be appointed underground mine manager by the coal mine operator for the mine.

(4) The site senior executive must appoint an alternate underground mine manager if the mine is to be managed in accordance with a commute system.

Maximum penalty—400 penalty units.

(5) The coal mine operator or site senior executive must not appoint a person as an underground mine manager unless the person has both of the following board qualifications—

(a) a first class certificate of competency for an underground mine;

(b) the practising certificate required by the board of examiners to be held by a person holding the board qualification mentioned in paragraph (a).

Maximum penalty—400 penalty units.

(6) A person must not give a direction to the underground mine manager about a technical matter in relation to the underground mine unless the person giving the direction is the holder of a first class certificate of competency for an underground coal mine.

Maximum penalty—200 penalty units.

(7) A person must not give a direction to the underground mine manager that may adversely affect safety and health at the underground mine.

Maximum penalty—200 penalty units.

(8) The underground mine manager must appoint a person holding both of the following board qualifications to be responsible for the control and management of underground activities when the manager is not in attendance at the mine—

(a) a first or second class certificate of competency or a deputy's certificate of competency for an underground mine;

(b) the practising certificate required by the board of examiners to be held by a person holding the board qualification mentioned in paragraph (a).

Maximum penalty—200 penalty units.

(9) The underground mine manager must appoint a person holding both of the following board qualifications to have control of activities in 1 or more explosion risk zones—

(a) a first- or second-class certificate of competency or a deputy's certificate of competency for an underground mine;

(b) the practising certificate required by the board of examiners to be held by a person holding the board qualification mentioned in paragraph (a).

Maximum penalty—200 penalty units.

(10) The underground mine manager must appoint a person, holding both of the following board qualifications, as electrical engineering manager to control and manage the electrical engineering activities of the mine—

(a) an electrical engineering manager's certificate of competency for an underground mine;

(b) the practising certificate required by the board of examiners to be held by a person holding the board qualification mentioned in paragraph (a).

Maximum penalty—200 penalty units.

(11) The underground mine manager must appoint a person, holding both of the following board qualifications, as mechanical engineering manager to control and manage the mechanical engineering activities of the mine—

(a) a mechanical engineering manager's certificate of competency for an underground mine;

(b) the practising certificate required by the board of examiners to be held by a person holding the board qualification mentioned in paragraph (a).

Maximum penalty—200 penalty units.

(12) The underground mine manager may appoint more than 1 person as the electrical engineering manager under subsection (10), or the mechanical engineering manager under subsection (11), to assume the duties of the manager at different times.

(13) A coal mine operator or site senior executive may appoint a person as underground mine manager for more than 1 mine at the same time only with the written approval of the chief inspector.

Maximum penalty—200 penalty units.

(14) The coal mine operator for the underground mine must ensure that a site senior executive required to appoint a person under subsection (2) or (4), or an underground mine manager required to appoint a person under subsection (8), (9), (10) or (11), appoints a person under the subsection only if the person is an employee of—

(a) for an appointment under subsection (2), (4) or (8)—

(i) the coal mine operator; or

(ii) an associated entity of the coal mine operator; or

(iii) an entity that employs or otherwise engages 80% or more of the coal mine workers at the coal mine; or

(b) for an appointment under subsection (9), (10) or (11)—

(i) the coal mine operator; or

(ii) an entity that employs or otherwise engages 80% or more of the coal mine workers at the coal mine.

Maximum penalty—500 penalty units.

200 Site not to be interfered with without permission

(1) A person must not interfere with a place at a coal mine that is the site of a serious accident or high potential incident of a type prescribed by regulation, without the permission of an inspector.

Maximum penalty—200 penalty units.

(2) Permission under subsection (1) must not be unreasonably withheld.

(3) For this division, action taken to save life or prevent further injury at a place is not interference with the place.

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16A Supervisor's responsibilities

A supervisor for a coal mine has the following responsibilities—

- (a) ensuring each mine worker for whom the supervisor has responsibilities is given the supervision and help mentioned in [sections 85A\(2\)](#) and [85B](#);
- (b) observing each coal mine worker for whom the supervisor has responsibilities perform tasks;
- (c) carrying out inspections of workplaces at the coal mine as required under the coal mine's safety and health management system;
- (d) recording reports of, and findings and recommendations resulting from, inspections mentioned in paragraph (c) in the mine record;
- (e) implementing and monitoring hazard controls and risk controls as required under the coal mine's safety and health management system.

17 Qualifications for appointment as electrical engineering manager

A person may appoint a person as an electrical engineering manager only if the person has the recognised competencies for controlling and managing the mine's electrical engineering activities.

18 Duties of electrical engineering manager

The duties of an electrical engineering manager include controlling and managing the following at the mine—

- (a) the design of electrical installations;
- (b) the installation and maintenance of electrical equipment and electrical installations;
- (c) electrical work;
- (d) work carried out close to electrical installations.

19 Duty to give notice of proposed introduction of electricity

Before electricity is introduced at a coal mine for use in coal mining operations, or after electricity is permanently disconnected, the site senior executive for the coal mine must notify an inspector of the proposed introduction, or disconnection.

20 Competencies of persons carrying out work on electrical equipment

(1) Subject to subsection (2), only a person who has the recognised competency for repairing, overhauling and maintaining electrical equipment for explosive atmospheres may carry out the following work at a coal mine—

- (a) electrical work in an ERZ;
- (b) work on electrical equipment used in an ERZ;
- (c) work on extra low voltage systems associated with explosion-protected equipment and installations;
- (d) work on explosion-protected electrical equipment installed at a surface mine;
- (e) electrical work in an area, at a surface mine, that a risk assessment shows has an explosion risk.

(2) Only a person who has the recognised competency for repairing, overhauling and maintaining electrical equipment for mobile plant used in underground mines may carry out work on extra low voltage systems associated with mobile plant for an underground mine.

21 Access to exposed electrical conductors

- (1) A coal mine must have a standard operating procedure for safely accessing exposed electrical conductors at the mine.
- (2) The procedure must provide for minimising the risk of energising exposed electrical conductors on which work is to be carried out, including, for example, by using—
- (a) personal locks; or
 - (b) danger tags; or
 - (c) personal locks and danger tags.
- (3) The procedure must also provide that before work is carried out on an exposed electrical conductor at above extra low voltage the conductor must be—
- (a) positively isolated from the electricity source; and
 - (b) tested for zero potential; and
 - (c) if the conductor is a high voltage conductor, earthed.
- (4) Subsection (5) applies only to a surface mine and only if—
- (a) it is not practicable to positively isolate the conductor from the electricity source; and
 - (b) access to the conductor is required or work in close proximity to the conductor is required to be carried out.
- (5) The procedure must also provide that—
- (a) a risk assessment, including consideration of high fault currents, must be carried out before the work is carried out; and
 - (b) if the risk assessment shows an unacceptable level of risk in carrying out the work with the conductor energised, before the work is carried out, the conductor must be—
 - (i) positively isolated from the electricity source; and
 - (ii) tested for zero potential; and
 - (iii) if the conductor is a high voltage conductor, earthed.
- (6) In this section—
- close proximity**, to a conductor, means—
- (a) for a person—
 - (i) for a high voltage conductor—within the exclusion zone for the person for the conductor under the [Electrical Safety Regulation 2013](#); or
 - (ii) for a low voltage conductor—within reach of the conductor; or
 - (b) for plant being used for work near an overhead power line—within the minimum clearances required under AS/NZS 3007:2013.

22 Locating and dealing with electrical faults

- (1) A coal mine must have a standard operating procedure for re-energising an electrical circuit at the mine that has been switched off by a circuit protection device.
- (2) A person may use electrical energy, other than electrical energy generated from appropriately designed test instruments, for locating, isolating or clearing only the following electrical faults—
- (a) an overload of an electric motor;
 - (b) an earth leakage trip associated with an impedance earth circuit;
 - (c) a 30mA earth leakage trip;
 - (d) an earth continuity trip.

23 Design, installation and maintenance

The site senior executive for a coal mine must ensure the design, installation and maintenance of electrical equipment and installations at the mine provide for the following—

- (a) reliable circuit interruption, under fault conditions, at all points in the mine's electrical distribution system;
- (b) each earthing system at the mine to be of sufficiently low impedance to ensure reliable operation of all electrical protection systems and devices and adequate protection against indirect contact;
- (c) limiting prospective touch voltage, under fault conditions, to within acceptable limits;
- (d) minimising potential impacts from voltage rise caused by lightning strike, static electricity, voltage surges and other transient voltages to within acceptable limits;
- (e) preventing a person inadvertently contacting live parts of equipment or an installation exceeding extra low voltage.

Example for paragraph (e)—

The design, installation and maintenance may provide for insulation or an enclosure, shield, interlocking device or automatic disconnection device for the equipment or installation.

24 Isolators for equipment driven by electricity

- (1) A coal mine's electrical engineering manager must ensure the mine has a full current isolator for equipment driven by electricity at the mine.
- (2) The electrical engineering manager must ensure—
 - (a) the isolator is—
 - (i) clearly identified as the isolator for the equipment; and
 - (ii) situated in a location that is easily accessible by a person working on the equipment; and
 - (b) the equipment is clearly identified as being supplied with electricity from the isolator.

Example of how the isolator or equipment may be identified for subsection (2)—

by marking the isolator or equipment, or placing a notice near the isolator or equipment

25 Standard for electrical control systems

A coal mine's electrical engineering manager must ensure—

- (a) each electrical control system at the mine operates safely under all operating conditions, including electricity supply instability or failure; and
- (b) if the system suffers a fault or fails, all emergency stopping systems and safety alarms at the mine remain effective.

26 Control circuits

- (1) A coal mine's electrical engineering manager must ensure a control circuit at the mine run externally to an enclosure from which the circuit originates—
 - (a) has a voltage of not more than a nominal voltage of 55V to earth; or
 - (b) is protected by earth leakage current protection of not more than 30mA sensitivity; or
 - (c) minimises the risk of direct contact and limits prospective touch voltage to within acceptable limits.
- (3) An underground mine's electrical engineering manager must ensure a control circuit for the mine is explosion protected, unless the whole circuit is installed and operated in a NERZ.

27 Modification of electrical control systems

A coal mine's safety and health management system must—

- (a) provide for the security and maintenance of the mine's electrical control system software and control circuits; and
- (b) control modification of the software and circuits; and
- (c) provide for records to be kept of any modifications.

28 Electrical protection for power outlets for low voltage electrical equipment

(1) A coal mine's electrical engineering manager must ensure a single-phase power outlet for low voltage electrical equipment at the mine has—

- (a) earth leakage current protection of not more than 30mA sensitivity; or
- (b) a device preventing its use by portable electrical equipment.

(2) Subsection (1) does not apply if the power outlet is the only outlet in an isolated electricity supply.

(3) In this section—

isolated electricity supply means an electricity supply—

- (a) other than the main electricity supply, that is totally insulated from conductive material; or
- (b) that is—
 - (i) supplied with electricity from the main electricity supply; and
 - (ii) totally isolated from the main electricity supply; and
 - (iii) totally insulated from earth.

29 Operating times and tripping current for circuit protection devices

(1) A coal mine's electrical engineering manager must ensure each circuit protection device at the mine has an appropriate operating time and tripping current, having regard to a risk assessment of the operation of the part of the electrical installation the device protects.

(2) For a surface mine, the operating time may allow for a delay in cutting off the electricity supply if—

- (a) the device is installed to protect a person operating equipment; and
- (b) it is appropriate to allow the person to bring the equipment to rest before the electricity supply is cut; and
- (c) the person is immediately given a clear warning of the fault.

30 Changing electrical protection settings

(1) A person must not change an electrical protection setting for a circuit protection device at a coal mine, unless the person is authorised to make the change by the electrical engineering manager for the mine.

(2) In this section—

electrical protection setting, for a circuit protection device, means the current, voltage or operating time at which the device is set to trip when the device detects an electrical fault in the circuit.

31 Unearthed electrical installations

(1) A coal mine's electrical engineering manager must ensure an unearthed electrical installation at the mine has earth fault detection and control.

(2) Subsection (1) does not apply to a single-phase 240V generator of less than 25kW capacity at a surface mine.

32 Drawings of electrical installations

- (1) The site senior executive for a coal mine must ensure up-to-date drawings of the mine's electrical installations are available for use at the mine when electrical equipment and electrical installations are commissioned or modified.
- (2) The drawings must be updated as soon as practicable, but not later than 8 days, after—
 - (a) the commissioning of any electrical equipment or electrical installation at the mine; or
 - (b) any major modification to electrical equipment or an electrical installation at the mine.
- (3) The drawings must be kept in a location that is easily accessible by each coal mine worker authorised by an electrical supervisor or the coal mine's electrical engineering manager to access the drawings.

33 Plan of coal mine's communication system and main electrical installation

- (1) The site senior executive for a coal mine must ensure the mine has a plan of the mine's main electrical installation showing the following—
 - (a) the location of each main electricity reticulation line;
 - (b) the location, rating, identifying label and purpose of each main isolator, substation and high voltage switchboard;
 - (c) any information required to perform switching programs;
 - (d) for a surface mine—the general location of each item of high voltage mobile plant supplied with electricity by a trailing cable.
- (2) The site senior executive for an underground mine must also ensure the mine has a plan of its communication system showing the location of each fixed communication device at the mine.
- (3) For an underground mine, the scale of each plan under this section must be 1:5000.
- (4) A plan under this section must be updated as soon as practicable, but not later than 8 days after—
 - (a) for a plan of the communication system—the installation or removal of a communication device; or
 - (b) for a plan of the main electrical installation—the relocation, removal or installation of a thing mentioned in subsection (1)(a), (b) or (d).
- (5) The site senior executive for a coal mine must ensure each

plan is kept in a location at the mine that is easily accessible by each coal mine worker authorised by an electrical supervisor or the mine's electrical engineering manager to access the plan.

34 Records

(1) A coal mine's electrical engineering manager or, if there is no electrical engineering manager for the mine, the site senior executive for the mine, must ensure a record of the following matters about the mine's electrical activities, equipment and installations is kept at the mine—

(a) for each electrical installation—

(i) its electrical protection settings, fault levels, and cable and conductor sizes and switchgear ratings, including circuit interrupting capacity; and

(ii) its commissioning dates and results; and

(iii) its monitoring and test dates and results;

(b) for each item of electrical equipment—

(i) its commissioning dates and results; and

(ii) its monitoring and test dates and results;

(c) for each item of electrical safety equipment used—

(i) its description; and

(ii) its test dates and results;

(d) the name of each person appointed—

(i) to control and manage the mine's electrical engineering activities; or

(ii) to carry out electrical work and electrical activities;

(e) the name of each person authorised to enter an electrical operating area;

(f) the activity each person mentioned in paragraph (e) is authorised to carry out in the operating area.

(2) In this section—

electrical operating area means an area or enclosure in which electrical equipment or an electrical installation is housed.

electrical safety equipment includes an instrument for testing electrical equipment, a ladder, a safety harness, an insulating tool and insulating gloves.

35 General

- (1) A coal mine's safety and health management system must provide for managing emergencies at the mine.
- (2) The system must provide for the following—
 - (a) identifying, by risk assessment, potential emergency situations;
 - (b) minimising risks associated with potential emergency situations;
 - (c) carrying out aided rescue and self-escape of persons from the mine in an emergency;
 - (d) carrying out emergency exercises, including testing the effectiveness of emergency management procedures and the readiness and fitness of equipment for use in an emergency;
 - (e) auditing and reviewing the emergency exercises;
 - (f) if the mine is a surface mine—involving an open-cut examiner for the mine in—
 - (i) developing and testing the emergency management procedures for activities, including mining activities, in and around the surface excavation at the mine; and
 - (ii) auditing the documentation for the procedures;
 - (g) if the mine is an underground mine—involving an ERZ controller for the mine in—
 - (i) developing and testing the emergency management procedures for explosion risk zones; and
 - (ii) auditing the documentation for the procedures.

37 Fire

- (1) A coal mine's safety and health management system must provide for the following at the mine—
 - (a) fire prevention and control;
 - (b) an effective fire fighting capability;
 - (c) the safety of persons fighting fires;
 - (d) a risk assessment to identify all potential fire hazards at the mine.
- (2) The system must also provide for the following—
 - (a) the availability at the mine, at all times, of equipment that is appropriate and sufficient to extinguish any potential fire identified by the risk assessment;
 - (b) the location of portable fire extinguishers on or near equipment and installations identified as potential fire hazards by the risk assessment;
 - (c) the compatibility, throughout the mine, of all fire fighting equipment.
- (3) The coal mine must have a standard operating procedure for action to be taken when a fire is discovered at the mine.

38 First aid

- (1) A coal mine's safety and health management system must provide for first aid at the mine.
- (2) The system must provide for the following—
 - (a) the supply and placement of first aid supplies and equipment, including supplies and equipment required for emergencies, based on a risk assessment;
 - (b) appropriate first aid training, including refresher training, emphasising the safety of the person giving first aid as well as the safety of the person receiving first aid;
 - (c) the availability of a sufficient number of persons competent in giving first aid.
- (3) For accidents involving electricity, the mine must have a standard operating procedure for the following—
 - (a) releasing a person from an energised low and high voltage conductor;
 - (b) resuscitating a person in an electrical environment;
 - (c) managing a person's flash burn injuries.
- (3) For accidents involving electricity, the mine must have a standard operating procedure for the following—

- (a) releasing a person from an energised low and high voltage conductor;
- (b) resuscitating a person in an electrical environment;
- (c) managing a person's flash burn injuries.

68 Modifying plant

- (1) A coal mine must have a standard operating procedure for modifying fixed and mobile plant.
- (2) A coal mine's safety and health management system must provide for the following—
 - (a) recording modifications made to the plant at the mine;
 - (b) updating drawings of the plant held at the mine to include the modifications;
 - (c) assessing and managing risk associated with the modifications.

69 Pre-start warning

- (1) This section applies if starting fixed or mobile plant at a coal mine is likely to cause a hazard to a person near the plant.
- (2) The mine's safety and health management system must provide for the plant to be fitted with a device that sounds a warning before the plant is started.

78 Isolating and tagging procedures

- (1) A coal mine must have a standard operating procedure for the following—
 - (a) controlling the risk of an unplanned release of energy from plant, including positively isolating the energy source;
 - (b) if an electrical or mechanical energy source is positively isolated—testing for zero potential;
 - (c) taking plant out of service;
 - (d) returning plant to service.
- (2) Without limiting subsection (1), the standard operating procedure may provide for the use of danger, isolation, operational, out of service, personal and restriction tags for particular circumstances.
- (3) If the safety or health of a person is directly affected by the operation or non-operation of plant, the procedure must also provide for the person to personally control, by attaching a danger tag or lock to the plant, the plant's change in status from non-operational to operational.
- (4) For subsection (1)(a), the method for positively isolating the energy source for plant provided for in the standard operating procedure must—
 - (a) for plant that is electrical equipment—be a manually initiated operation that—
 - (i) isolates all active power conductors of the plant being isolated from the electricity supply; and
 - (ii) prevents unintended re-energisation, including re-energisation through inadvertent operation of the plant or component failure; and
 - (b) for other plant—be the operation of a manually operated device that—
 - (i) is installed in the energy supply for the plant and, when operated, isolates the plant from its energy source; and
 - (ii) requires a manually initiated operation for the supply of energy to the plant to be resumed; and
 - (c) require the isolation device be clearly marked as being the positive means of isolating the plant.
- (5) In this section—
isolation device means—
 - (a) for plant that is electrical equipment—the device used manually to initiate the operation mentioned in subsection (4)(a); or
 - (b) for other plant—the device mentioned in subsection (4)(b).

79 Equipment used for maintaining plant

- (1) A coal mine must have a standard operating procedure for electrical and mechanical equipment used for inspecting, testing and maintaining the safe operation of plant at the mine.
- (2) The procedure must provide for the following—
 - (a) selecting equipment that is fit for its intended purpose;
 - (b) using the equipment safely;
 - (c) maintaining the equipment in accordance with its specification;
 - (d) testing the equipment at appropriate intervals and labelling the equipment to show when the equipment was last tested.

81 Action to be taken if certified equipment does not meet certification requirements or is likely to create an unacceptable level of risk

- (1) Subsection (2) applies if a nationally accredited testing station becomes aware that—
 - (a) equipment certified by the testing station should not have been certified because the equipment did not meet the requirements for certification; or
 - (b) the requirements for certification change in a way that is likely to create an unacceptable level of risk if equipment that has been certified by the testing station is used for the purpose for which the equipment was certified; or
 - (c) the certification of equipment is amended, suspended or cancelled.
- (2) The testing station must give immediate notice of the fact to—
 - (a) the manufacturer or supplier of the equipment who obtained the certification; and
 - (b) the chief inspector.

85A Time and resources for carrying out tasks

- (1) The site senior executive for a coal mine must ensure time is allocated, and the mine's resources are distributed, to enable each coal mine worker at the mine to carry out the worker's tasks without creating an unacceptable level of risk.
- (2) Without limiting subsection (1), the site senior executive must ensure the coal mine worker is given the supervision, and help from other competent persons, necessary to achieve an acceptable level of risk.

85B Supervising workers

- (1) Without limiting [section 85A](#)(2), the site senior executive for a coal mine must ensure the coal mine's activities and coal mine workers are supervised to the extent necessary to ensure each worker—
 - (a) is not likely to be exposed to conditions beyond the worker's capabilities; and
 - (b) is not likely to be affected by the conditions in which the worker is working in a way that adversely affects the worker's fitness to perform critical tasks; and
 - (c) has the resources necessary to undertake the work without being exposed to an unacceptable level of risk; and

(d) is working within the limits of the worker's fitness and competence; and

(e) complies with the worker's safety and health obligations.

(2) The supervision must include communication, including direct contact, between the supervisor and the coal mine worker at appropriate intervals.

86 Air conditioning units

A coal mine's safety and health management system must provide for maintaining and cleaning air conditioning units used at the mine.

88 Cutting, drilling or excavating near a concealed service

(1) A coal mine's safety and health management system must provide for controlling risks from cutting or drilling into a building or structure, or excavating ground, at the mine where there may be a concealed service.

(2) The system must provide for the following to minimise the uncontrolled release of non-electrical or electrical energy from a concealed service—

(a) locating the energy source and placing a mark, describing the source, on or near the source;

(b) isolating the energy source;

(c) testing the energy source and proving the source de-energised.

(3) The site senior executive for the coal mine must provide sufficient equipment for the matters mentioned in subsection (2)(a) to (c).

112 Live testing electrical equipment

(1) A surface mine must have a standard operating procedure for live testing electrical equipment at the mine.

(2) The procedure must—

(a) state—

(i) what equipment may be live tested; and

(ii) what instruments may be used for the testing; and

(b) provide for the following—

(i) isolating areas where equipment is being live tested or test run;

(ii) using personal protective equipment;

(iii) using high voltage test equipment and discharging stored energy after the test;

(iv) dealing with high prospective fault currents;

(v) other precautions to be taken for live testing.

113 Electrical protection for electrical distribution system

(1) The site senior executive for a surface mine must ensure an earthed electrical distribution system at the mine has earth leakage protection.

(2) Subsection (1) does not apply if—

(a) persons at the mine are protected against indirect contact with the system; or

(b) for an overhead power line—

(i) mobile plant is not operated close to the line; and

(ii) a risk assessment shows earth leakage protection would only minimally reduce the risk.

114 Electrical protection for mobile or transportable equipment

- (1) This section applies to mobile or transportable equipment used at a surface mine and supplied with electricity by a reeling or trailing cable.
- (2) If the equipment does not have a local earthing system, the site senior executive for the mine must ensure the cable has an earth continuity monitoring system that cuts off the electricity supply immediately after an earth continuity fault is detected in the cable or equipment.
- (3) Subsection (2) does not apply if the cable—
 - (a) is a low voltage cable; and
 - (b) has earth leakage current protection of not more than 30mA sensitivity; and
 - (c) has individually screened cores; and
 - (d) has no joining plugs or sockets.
- (4) The site senior executive must ensure the electricity source for the equipment has an earth fault current limitation device that limits prospective touch voltage to within acceptable limits.
- (5) Subsection (4) does not apply if the equipment is low voltage equipment and persons at the mine are otherwise protected adequately against injury from indirect contact with electricity.

127 Plans of surface services

- (1) The site senior executive for a surface mine must ensure current plans identifying and showing the location of surface services, including buried services, are kept in a location at the mine that is easily accessible by each coal mine worker at the mine.
- (2) In this section—

buried service includes electricity conducted through an underground electrical cable and another service carried in an underground pipe.

surface service includes the following—

 - (a) a communication cable;
 - (b) compressed air storage and reticulation;
 - (c) diesel storage and reticulation;
 - (d) an electrical cable;
 - (e) an emergency access or road;
 - (f) a fire fighting facility;
 - (g) a first aid facility;
 - (h) gas storage and reticulation;
 - (i) rescue equipment;
 - (j) a sewerage facility;
 - (k) a water storage pipeline or drain.

132 Using and relocating operator-attended dredges

- (1) A surface mine must have a standard operating procedure for using and relocating operator-attended dredges at the mine.
- (2) The procedure must provide for the following—
 - (a) safe access to a floating pontoon;
 - (b) restricting access to areas identified in a risk assessment as high risk areas;
 - (c) controlling electrical trailing cables connected to the dredge;
 - (d) the safety of persons near winch ropes;
 - (e) using a bilge pump, including, in particular, ways to minimise the risk of reverse siphoning;
 - (f) checking and recording the dredge's freeboard;

- (g) maintaining and inspecting the dredge's hulls and parts essential for flotation;
- (h) controlling risk from dredging the face;
- (i) monitoring climatic conditions and securing the dredge during adverse climatic conditions.

135 Using plant near electricity

- (1) A surface mine must have a standard operating procedure for using fixed and mobile plant near electrical cables, overhead electrical conductors, or electrical equipment.
- (2) The procedure must provide for the following—
 - (a) recording power line heights at each location where a vehicle may pass;
 - (b) keeping a register of the height of vehicles and mobile plant used on site;
 - (c) fixing standard travelling routes at the mine;
 - (d) travelling other than on a standard travelling route;
 - (e) inspecting, and reporting on, travelling routes for plant not included in the register mentioned in paragraph (b);
 - (f) preventing incidents involving electrical contact, including, for example, by—
 - (i) using barriers, signs, indicators and alarms; and
 - (ii) requiring stated clearance distances between plant and electrical conductors; and
 - (iii) isolating electrical conductors.

137 Fire detection and suppression in tunnels

A surface mine's safety and health management system must provide for automatically detecting combustion and suppressing fires in tunnels beneath stockpiles.

144 Lighting

- A surface mine's safety and health management system must provide for the following—
 - (a) adequate lighting, having regard to shadow, contrast and glare, in each area of the mine in which work is to be carried out so the work can be carried out safely;
 - (b) emergency lighting to ensure that persons in a building or enclosure at the mine can safely exit if the normal lighting fails.

146 Trenches

- (1) A surface mine's safety and health management system must provide for protecting persons from risks from trenches at the mine.
- (2) The system must provide for the following—
 - (a) the use of barricades;
 - (b) ground support;
 - (c) water ingress and hazardous atmosphere in the trench.

152 Limit to external surface temperature of equipment used underground

Unless otherwise allowed under [part 8, division 5](#), a person must not use equipment underground if its external surface temperature is more than 150°C.

161 Types of cap lamps

- An underground mine's safety and health management system must provide for the following—
 - (a) cap lamps for use at the mine to be of a certified type and explosion protection category Ex ia, Ex s or Ex l;

(b)a record to be kept of the following for each cap lamp type in use at the mine—

- (i)its design and construction specifications;
- (ii)the manufacturer's maintenance and testing recommendations.

162Facilities and equipment for cap lamps

The site senior executive for an underground mine must ensure—

- (a)the mine has the following facilities and equipment for cap lamps—
 - (i)a lamp room;
 - (ii)battery charging stations;
 - (iii)cap lamp battery charging equipment;
 - (iv)the correct tools and testing equipment for repairing and maintaining cap lamps; and
- (b)each battery charging station is adequately ventilated to remove gases generated by the charging.

163 Maintaining and testing cap lamps

The underground mine manager for an underground mine must ensure—

- (a)each cap lamp used at the mine is operated for the minimum period stated in the standard against which the cap lamp is certified immediately before the cap lamp is tested; and
- (b)the cap lamp is used again only if the test shows its performance is at least 60% of the performance stated in the standard; and
- (c)a record is kept of results of maintenance and testing of cap lamps in use at the mine.

176 Telephonic communication

- (1) The site senior executive for an underground mine must ensure the mine's telephonic communication system complies with this section and has an adequate failsafe, or backup, power supply for the system.
- (2) The system's electrical components installed underground must be suitable for use in an ERZ0, unless the components are installed in a drift or shaft being driven from the surface in material other than coal.
- (3) The system must provide for effective telephonic communication to and from the following places at the mine—
 - (a) each entrance underground, and on the surface, through which a person may enter into a shaft or other excavation used for ingress to or egress from the mine workings;
 - (b) each underground battery charging station;
 - (c) each underground workshop;
 - (d) each underground crib room;
 - (e) a place close to each switchgear used to isolate or control sections of the underground main electricity distribution system;
 - (f) a place close to each underground conveyor belt drivehead;
 - (g) a place close to each underground loading or transfer point on the conveyor belt system;
 - (h) each emergency refuge chamber;
 - (i) a place in each inspection district in the mine not otherwise mentioned in paragraphs (a) to (h).

178 Communication by notices and plans

(1)The underground mine manager for an underground mine must ensure the mine has noticeboards for displaying safety and health notices and plans—

- (a)at the surface of the mine where coal mine workers assemble before going underground; and
- (b)in each underground crib room.

(2)The underground mine manager must ensure the safety and health notices mentioned in subsection (3) are displayed at—

- (a)each place mentioned in subsection (1); and
- (b)each place containing fixed electrical equipment other than the following—
 - (i)cables;
 - (ii)switchgear associated with portable equipment;
 - (iii)isolated remote control switches;
 - (iv)lighting;
 - (v)communications and signalling equipment.

(3)For subsection (2), the safety and health notices are notices about the following—

- (a)directions for resuscitating persons suffering from electric shock;
- (b)directions for the procedure in case of fire;
- (c)a direction prohibiting a person from handling, operating or interfering with electrical equipment at the place unless the person—
 - (i)is authorised to take the action; or
 - (ii)is taking action to cut off the electricity supply in an emergency.

179 Safety and health management system

An underground mine's safety and health management system must provide for the following—

- (a)the safe and secure location of the mine's electrical equipment and installations;
- (b)the design and operation of liquid-filled electrical equipment.

181 ERZ0

(1)The site senior executive for an underground mine must ensure electrical equipment installed or operated in an ERZ0 at the mine—

- (a)is suitable for use in an underground mine; and
- (b)is certified as having explosion protection category Ex ia, Ex s or Ex l.

(2)Subsection (1)(b) does not apply to electrical cables that are part of an Ex ia, Ex s or Ex l circuit.

(3)Also, subsection (1)(b) does not apply in relation to electrical equipment operated in a mine, or part of a mine, that is an ERZ0 if—

- (a)the mine, or part of the mine, has been sealed and does not have an explosive atmosphere; and
- (b)the equipment is operated for the purpose of assessing the risk of—

- (i)persons re-entering the mine or part of the mine; or
- (ii)ventilating the mine or part of the mine; and

(c)a risk assessment is carried out before the equipment is operated in the mine, or part of the mine, to—

- (i)identify the oxygen nose point for the atmosphere in the mine or part of the mine; and
- (ii)identify hazards involved in operating the equipment in the mine or part of the mine; and
- (iii)identify ways of effectively controlling the hazards; and

(d)the atmosphere around the equipment is continuously monitored using a gas monitoring device to detect the general body concentration of oxygen and methane; and

(e)both of the following actions are taken if the general body concentration of oxygen around the equipment exceeds 60% of the oxygen nose point identified in the risk assessment—

- (i) the electrical supply to the equipment is immediately switched off;
 - (ii) the equipment is immediately withdrawn to a safe location; and
 - (f) any hazards identified in the risk assessment can be effectively controlled; and
 - (g) any ways identified in the risk assessment of effectively controlling the hazards are implemented.
- (4) The gas monitoring device mentioned in subsection (3)(d) must automatically activate an audible or visible alarm when the general body concentration of oxygen exceeds 60% of the oxygen nose point identified in the risk assessment.

182 ERZ1

- (1) The site senior executive for an underground mine must ensure fixed, mobile and transportable electrical equipment installed or operated in an ERZ1 at the mine—
- (a) is suitable for use in an underground mine; and
 - (b) is certified as having explosion protection.
- (2) Subsection (1) does not apply to electrical equipment associated with hot work or live testing.
- (3) Subsection (1)(b) does not apply to a power transformer, traction battery or electrical cable.

183 NERZ

- (1) The site senior executive for an underground mine must ensure fixed, mobile and transportable electrical equipment installed or operated in a NERZ at the mine—
- (a) is suitable for use in an underground mine; and
 - (b) either—
 - (i) is certified as having explosion protection; or
 - (ii) has a degree of protection of, or equivalent to, at least IP55 under AS 60529—2004.
- (2) Subsection (1) does not apply to electrical equipment—
- (a) associated with hot work or live testing; or
 - (b) installed in a pressurised room.
- (2A) Subsection (1)(b) does not apply to electrical cables.
- (3) In this section—

pressurised means pressurised within the meaning of AS 2380.4—1994 ‘Electrical equipment for explosive atmospheres—Explosion protection techniques—Pressurized rooms or pressurized enclosures’.

184 Enclosures for switchgear

The site senior executive for an underground mine must ensure each enclosure containing switchgear at the mine—

- (a) is suitably rated for the prospective electrical fault level at the switchgear; and
- (b) has adequate arc fault control.

185 Record of suitability of electrical equipment and installations

- (1) The electrical engineering manager for an underground mine must ensure documentation showing the use for which the mine’s electrical equipment and installations is suitable is kept at the mine.
- (2) For electrical equipment mentioned in [section 181](#), [182](#) or [183](#), the documentation must include the certification mentioned in the section.

186 Live testing in an ERZ

- (1) A person must not live test electrical equipment or installations, including intrinsically safe electrical equipment and installations, in an ERZ0 at an underground mine.
- (2) An underground mine's safety and health management system must provide for—
 - (a) live testing electrical equipment and installations in an ERZ1; and
 - (b) limiting the live testing to extra low voltage and low voltage; and
 - (c) notice of the live testing, other than testing of intrinsically safe electrical equipment and installations, to be given to an inspector within 7 days after the live testing is carried out.

187 Live testing in a NERZ

- (1) An underground mine must have a standard operating procedure for live testing electrical equipment and installations in a NERZ.
- (2) The procedure must provide for the following—
 - (a) limiting live testing to extra low voltage and low voltage electrical equipment and installations;
 - (b) isolating areas where equipment is being live tested;
 - (c) the type of test instruments to be used;
 - (d) using high voltage test equipment and discharging stored energy after testing;
 - (e) using personal protective equipment;
 - (f) dealing with high prospective fault currents.

188 Selecting, installing and using electrical cables and accessories

An underground mine's safety and health management system must provide for selecting, installing and using electrical cables and electrical cable accessories for use at the mine.

189 Repairing and testing reeling, trailing and feeder cables

- (1) An underground mine's safety and health management system must provide for repairing and testing reeling, trailing and feeder cables.
- (2) A person must not use a reeling, trailing or feeder cable at an underground mine if the cable—
 - (a) has been found to be defective; and
 - (b) has not been effectively repaired and tested under the system.
- (3) Subsection (2) does not apply if—
 - (a) only the outer sheath of the cable is damaged; and
 - (b) before the cable is used, the sheath is repaired in a way that satisfies a competent person that the cable can be used without creating an unacceptable level of risk.
- (4) In this section—

competent person means a person who is competent in assessing the level of risk from a damaged cable.

190 Circuit separation

The electrical engineering manager for an underground mine must ensure each electrical circuit supplying the mine's underground electrical distribution system, ventilation system or winding or haulage system is designed so an electrical fault on 1 circuit will not affect another circuit.

191 Earth fault current limitation

(1) The electrical engineering manager for an underground mine must ensure each electrical circuit at the mine has an earth fault current limitation of not more than—

- (a) for a circuit supplying equipment and cables in an ERZ1 at the mine—5A; or
- (b) for a high voltage circuit not mentioned in paragraph (a)—10A; or
- (c) for a low voltage circuit not mentioned in paragraph (a)—5A.

(2) Subsection (1) does not apply to the following—

- (a) an intrinsically safe circuit;
- (b) a circuit operating at not more than 55V to earth and fully contained in a NERZ or an explosion-protected enclosure;
- (c) a circuit isolated from earth.

(3) The electrical engineering manager must ensure—

- (a) the neutral point of a transformer secondary winding for an impedance earthed electrical system at the mine is distributed to only the neutral earthing impedance; and
- (b) each part of an impedance earthed, or insulated neutral, electrical system at the mine has a phase-to-earth insulation level equivalent to the phase-to-phase insulation level for the electrical system.

192 Earth leakage protection

(1) The electrical engineering manager for an underground mine must ensure each electrical circuit at the mine operating above 55V to earth has earth leakage protection with a rated tripping current of not more than—

- (a) for a circuit supplying equipment and cables in an ERZ1—500mA; or
- (b) for a high voltage circuit not mentioned in paragraph (a)—1A; or
- (c) for a low voltage circuit not mentioned in paragraph (a)—500mA.

(2) The electrical engineering manager must ensure the operating time ensures the device cuts off the electricity supply to equipment and cables in an ERZ1, or causes the supply to be

cut off, as soon as practicable after the device detects an electrical fault in the circuit.

(3) The electrical engineering manager must ensure each circuit supplying portable electrically operated handheld plant—

(a) operates at not more than 250V; and

(b) has earth leakage current protection of not more than 30mA sensitivity.

(4) The site senior executive for the mine must ensure—

(a) each circuit supplying a reeling, trailing or feeder cable has—

(i) earth continuity protection; and

(ii) a lockout earth fault current protection device; and

(b) if the device is for equipment used in an ERZ1—the device is intrinsically safe.

(5) Subsection (1) does not apply to the following—

(a) an intrinsically safe circuit;

(b) a circuit operating at not more than 55V to earth and fully contained in a NERZ or an explosion-protected enclosure;

(c) a circuit isolated from earth.

(6) In this section—

rated tripping current means the electrical current at which the circuit breaker is set to trip.

193 Testing earth leakage protection and earth continuity protection

The electrical engineering manager for an underground mine must ensure electrical equipment at the mine having an earth leakage protection device, or an earth continuity protection device, has an easily accessible external test button on the equipment for effectively testing the device.

194 Limitation on earthing conductor's current

(1) The electrical engineering manager for an underground mine must ensure an earthing conductor at the mine is not used to carry an electrical circuit's normal current.

(2) Subsection (1) does not apply if the conductor is used for earth continuity protection and as a lockout earth fault current protection device.

195 Earthing electrodes

- (1) The electrical engineering manager for an underground mine must ensure earthing electrodes installed for the underground electrical distribution system's connection to earth are located at the surface of the mine.
- (2) If the mine has more than 1 electrical distribution system, the mine's safety and health management system must provide for the prospective touch voltage at the mine to be limited to a level necessary to achieve an acceptable level of risk.
- (3) The electrical engineering manager must ensure the ohmic resistance of an earthing electrode assembly for the underground electrical distribution system is kept as low as practicable and does not exceed 1 ohm.

197 Battery-powered vehicles

- (1) An underground mine's safety and health management system must provide for selecting, constructing, installing and maintaining battery-powered vehicles used at the mine.
- (2) The site senior executive for the mine must ensure a non-explosion-protected battery-powered vehicle is used—
 - (a) only in a NERZ; and
 - (b) only if the vehicle's access into an ERZ0 or ERZ1 is prevented by a system that is either fail safe or includes multiple redundancy devices.

198 Traction battery powered vehicles

The site senior executive for an underground mine must ensure—

- (a) a vehicle powered by a traction battery and used at the mine is fitted with a detector for, and protection from, battery earth leakage current; and
- (b) the traction battery is adequately ventilated to prevent accumulations of explosive mixtures of hydrogen and air forming in the battery's immediate vicinity.

200 Switching and isolating electricity supply

- (1) The underground mine manager for an underground mine must ensure the mine has, at each place where persons enter underground from the surface—
 - (a) a suitable communication method for advising a person to switch off the electricity supply to underground electrical equipment and cables; or
 - (b) a way to switch off the electricity supply.
- (2) The electrical engineering manager must ensure a notice giving instructions for subsection (1)(a) and (b) is placed on the noticeboard—

- (a) at the surface of the mine where coal mine workers assemble before going underground; and
- (b) in each underground crib room.

201 Using fibre optics

An underground mine's safety and health management system must provide for the safe use of fibre optic equipment at the mine.

202 Using portable electrical equipment

- (1) An underground mine's safety and health management system must provide for using certified and uncertified portable electrical equipment, other than portable gas detectors, in an ERZ1.
- (2) The mine must have a standard operating procedure for using certified and uncertified portable electrical equipment, other than portable gas detectors, in a NERZ.

222 Gas monitoring system

- (1) An underground mine's safety and health management system must provide for a gas monitoring system complying with this section.
- (2) The gas monitoring system must provide for the following—
 - (a) continuous monitoring of the mine atmosphere at the return airway of each ventilation split, to detect methane, carbon monoxide, carbon dioxide and oxygen;
 - (b) automatically detecting or calculating the values and trends of the following—
 - (i) gas concentrations;
 - (ii) the ratio of carbon monoxide and oxygen deficiency (commonly known as Graham's ratio);
 - (iii) the ratio of carbon monoxide and carbon dioxide;
 - (iv) gas explosibility;
 - (c) automatically activating an alarm if a gas alarm level is exceeded;
 - (d) recording the values and trends mentioned in paragraph (b) and displaying the record—
 - (i) at the surface of the mine where the record can be easily accessed by coal mine workers; and
 - (ii) in a way that the record can be easily read by the workers;
 - (e) keeping the information on which the values and trends mentioned in paragraph (d) were based at the mine in a way that enables the information to be easily accessed and inspected.
- (3) The gas monitoring system must also provide for—

- (a) an alternative electricity supply to ensure the system continues to function if the normal electricity supply fails; and
- (b) electrical equipment installed and operated underground for the system to have the following explosion protection category—
 - (i) for equipment other than a gas detector head—Ex ia;
 - (ii) for a gas detector head—Ex ia or Ex s.

223 Monitoring and sampling mine atmosphere

(1) An underground mine's safety and health management system must provide for continuous monitoring of the mine atmosphere, using the mine's gas monitoring system, at the return airway of each ventilation split.

(1A) The safety and health management system must also provide for sampling of the mine atmosphere, using the mine's gas monitoring system, at each of the following places—

- (a) the return airway from each goaf area and each set of workings;
- (b) the return of each airway at the upcast shaft;
- (c) other places stated in the mine's principal hazard management plan for gas monitoring as places where gas monitoring must be carried out.

(1B) The safety and health management system must also provide for—

- (a) continuous monitoring, using the mine's gas monitoring system, to detect products of combustion in the mine atmosphere at the return side of each conveyor belt; and
- (b) when the products are detected, the automatic activation of an alarm located on the surface in a position that is generally, under observation to warn persons of the products' presence.

(2) The safety and health management system must also provide for a regularly updated plan to be kept at the mine showing the location of—

- (a) the sampling point for each of the places mentioned in subsections (1), (1A) and (1B); and
- (b) each of the mine's ventilation control devices designated under section 351(1).

(3) A person must not relocate equipment used for sampling without the mine's ventilation officer's authorisation.

224 Gas alarm levels

(1) An underground mine's principal hazard management plan for gas management must state the values and ratios for gas, mentioned in section 222(2)(b), that are gas alarm levels.

(2) The mine must have a standard operating procedure for changing the gas alarm levels, including recording the following details—

- (a) the nature of, and reason for, the change;
- (b) the date the change was made;
- (c) the name of the person who made the change.

225 Changing gas alarm level settings

A person must not change a gas alarm level setting for an underground mine without the mine's ventilation officer's authorisation.

226 Acknowledging alarms

(1) An underground mine must have a standard operating procedure for acknowledging alarms that are activated when gas alarm levels are exceeded.

(2) The procedure must provide for at least 1 person to be—

- (a) on the surface when a person is underground; and
- (b) authorised by the underground mine manager for the mine to acknowledge the alarms.

227 Portable gas detectors

The site senior executive for an underground mine must ensure—

(a) the mine has a sufficient number of portable gas detectors capable of detecting the presence of methane, carbon monoxide and oxygen in the mine atmosphere; and

(b) each portable gas detector used at the mine is certified as—

- (i) suitable for use in an underground mine; and
- (ii) having the following explosion protection category—

(A) for parts of the detector other than the gas detector head—Ex ia;

(B) for the gas detector head—Ex ia or Ex s; and

(iii) capable of detecting the type of gas for which the detector is intended to be used; and

(iv) being accurate and reliable.

228 Providing portable gas detectors

- (1) The underground mine manager for an underground mine must ensure each person appointed under section 60(8) or (9) of the Act for the mine is provided with a portable gas detector.
- (2) The underground mine manager may provide a portable gas detector to a person, other than a person appointed under section 60(8) or (9) of the Act, only if the person is—
 - (a) appointed by the underground mine manager to use the detector; and
 - (b) competent in its use.
- (3) A person provided with a portable gas detector for an underground mine must keep the detector in the person's possession or under the person's direct control while the detector is in use underground.

229 Fixed methane detectors

- (1) This section applies to a fixed methane detector at an underground mine that is—
 - (a) fitted to equipment; or
 - (b) a self-contained unit located at a particular place; or
 - (c) part of the gas monitoring system.
- (2) The site senior executive for the mine must ensure that if the detector malfunctions or fails the detector will automatically—
 - (a) cause the equipment, or part of the equipment, the detector is monitoring to shut down; and

231 Auxiliary or booster fan

- (1) An auxiliary or booster fan must be protected by at least 1 methane detector to detect the general body concentration of methane at the fan.
- (2) For an auxiliary fan, the detector must be an automatic methane detector that trips the electricity supply to the fan when the concentration exceeds 2%.
- (3) For a booster fan, the detector must, when the concentration exceeds 1.25%, automatically activate an audible and visible alarm located in a place that allows the necessary action to be taken promptly.
- (4) If the detector protecting an auxiliary fan fails or is otherwise non-operational, the underground mine manager must ensure that, while the fan is operating, a person—
 - (a) continuously monitors the general body concentration of methane at the fan by using a portable methane detector that gives an audible and visible alarm when the concentration exceeds 1.25%; and

(b) disconnects the electricity supply to the fan when the concentration exceeds 1.25%.

(5) This section does not apply to an auxiliary or booster fan for a drift or shaft being driven from the surface in material other than coal.

232 Main exhausting fan

(1) The ventilating air passing through a main exhausting fan must be monitored by at least 1 automatic methane detector to detect the air's general body concentration of methane.

(2) The detector must automatically activate a visible alarm when the concentration exceeds the percentage stated in the mine's principal hazard management plan for ventilation as the percentage that must not be exceeded before the detector activates the alarm.

233 Coal cutter, continuous miner, tunnel boring and road heading machine

(1) A coal cutter, continuous miner, tunnel boring or road heading machine must be fitted with at least 1 automatic methane detector to detect the methane concentration near the cutters.

(2) If the machine is fitted with only 1 automatic methane detector, the detector must automatically—

(a) activate a visible alarm to warn the operator when the concentration exceeds 1%; and

(b) trip the electricity supply to the machine when the concentration exceeds 2%.

(3) If the machine is fitted with more than 1 automatic methane detector—

(a) a detector fitted to detect the methane concentration near the cutters must automatically—

(i) activate a visible alarm to warn the operator when the concentration exceeds 1%; and

(ii) trip the electricity supply to the cutters when the concentration exceeds 2%; and

(b) a detector fitted to detect the general body concentration of methane around the machine must automatically—

(i) activate a visible alarm to warn the operator when the concentration exceeds 1%; and

(ii) trip the electricity supply to the machine when the concentration exceeds 2%.

234 Longwall shearer

- (1) A longwall shearer must be fitted with at least 1 automatic methane detector to detect the general body concentration of methane around the machine.
- (2) The detector must automatically—
 - (a) activate a visible alarm to warn the operator when the concentration exceeds 1%; and
 - (b) trip the electricity supply to—
 - (i) the cutters when the concentration exceeds 1.25%; and
 - (ii) the machine when the concentration exceeds 2%.

234A Armoured face conveyor

- (1) An armoured face conveyor must be fitted with at least 1 automatic methane detector on each electrically-powered tailgate drive motor to detect the general body concentration of methane at the tailgate drive motor.
- (2) The detector must automatically—
 - (a) activate a visible alarm to warn the operator when the concentration exceeds 1%; and
 - (b) trip the electricity supply to the armoured face conveyor and the longwall shearer cutters when the concentration exceeds 2%.

Note—

See section 393.

235 Mobile bolting machine

- (1) A mobile bolting machine must be fitted with at least 1 automatic methane detector to detect the general body concentration of methane around the machine.
- (2) The detector must automatically—
 - (a) activate a visible alarm to warn the operator when the concentration exceeds 1%; and
 - (b) trip the electricity supply to the machine when the concentration exceeds 2%.

236 Explosion-protected electrically-powered loader

- (1) An explosion-protected electrically-powered loader must be fitted with at least 1 automatic methane detector to detect the general body concentration of methane around the machine.
- (2) The detector must automatically—
 - (a) activate a visible alarm to warn the operator when the concentration exceeds 1%; and
 - (b) trip the electricity supply to the loader when the concentration exceeds 2%.

237 Explosion-protected load-haul dump vehicle powered by a battery or internal combustion engine

(1) This section applies to an explosion-protected load-haul dump vehicle powered by a battery or an internal combustion engine.

(2) The vehicle must be fitted with at least 1 automatic methane detector to detect the general body concentration of methane around the vehicle.

(3) The detector must automatically—

(a) activate a visible alarm to warn the operator when the concentration exceeds 1%; and

(b) either—

(i) trip the electricity supply to the vehicle's electrical motors when the concentration exceeds 2%; or

(ii) stop the vehicle's internal combustion engine when the concentration exceeds 1.25%.

238 Other explosion-protected plant powered by battery or internal combustion engine

(1) Battery, or internal combustion engine, powered explosion-protected plant, other than a load-haul dump vehicle, must be fitted with at least 1 automatic methane detector to detect the general body concentration of methane around the plant.

(2) The detector must automatically—

(a) activate a visible alarm to warn the operator when the concentration exceeds 1%; and

(b) either—

(i) trip the electricity supply to the plant's electrical motors when the concentration exceeds 2%; or

(ii) stop the plant's internal combustion engine when the concentration exceeds 1.25%.

(3) This section does not apply to plant that is being operated—

(a) by a person using a portable methane detector that gives an audible and visible alarm when the concentration exceeds 1% and 1.25%; and

(b) in a location inspected periodically by a person using a portable methane detector.

(4) In this section—

periodically, for inspecting plant in a location in an ERZ1, means at least twice during a shift at approximately evenly spaced intervals.

240 Non-explosion-protected plant

(1) This section applies to non-explosion-protected plant, other than portable plant, powered by a battery or an internal combustion engine.

(2) The plant must be fitted with at least 1 automatic methane detector to detect the general body concentration of methane around the plant.

(3) The detector must automatically—

(a) activate—

(i) a visible alarm to warn the operator when the concentration exceeds 0.25%; and

(ii) an audible or visible alarm to warn the operator when the detector fails in service; and

(b) trip the electricity supply to the plant's electrical motors or stop its internal combustion engine—

(i) when the concentration exceeds 0.5%; and

(ii) within 3 minutes after the detector fails in service.

242 Intake airways

(1) At least 1 automatic methane detector must be located in each intake airway at the interface between—

(a) a NERZ and ERZ1; and

(b) 2 NERZs.

Example of interface between 2 NERZs—

the interface between subdivided parts of a NERZ

(2) A detector located at an interface between a NERZ and ERZ1 must—

(a) when the general body concentration of methane detected at the interface exceeds 0.25%—automatically activate a visible alarm; and

(b) when the general body concentration of methane detected at the interface exceeds 0.5%—automatically trip the electricity supply to non-intrinsically safe plant in—

(i) the ERZ1 and NERZ; or

(ii) if the NERZ has been subdivided—the ERZ1 and the subdivided part of the NERZ adjacent to the ERZ1.

(3) A detector located at the interface between a NERZ and an ERZ1 must be a self-contained unit or part of the gas monitoring system for the mine.

(4) A detector located at an interface between 2 NERZs must—

(a) automatically activate a visible alarm when the general body concentration of methane detected at the interface exceeds 0.25%; and

(b) if the NERZ has been subdivided—automatically trip

the electricity supply to non-intrinsically safe plant in the adjacent subdivided part when the general body concentration of methane detected at the interface exceeds 0.5%.

(5) The alarm mentioned in subsection (2)(a) or (4)(a) must be visible at the interface.

243A Longwall return airway generally

(1) This section applies—

(a) in relation to a longwall return airway; and

(b) in addition to the requirements under section 243.

(2) At least 1 automatic methane detector must be located—

(a) in the longwall return airway; and

(b) within 150m outbye of the corner of the tailgate block side rib line and the longwall coal face; and

(c) so as to measure the general body concentration of methane in the longwall return airway from—

(i) the longwall face; and

(ii) the goaf.

(3) The detector must automatically—

(a) activate a visible alarm when the general body concentration of methane detected in the return air exceeds the percentage stated in the mine's principal hazard management plan for ventilation as the percentage that must not be exceeded before the detector activates the alarm; and

(b) trip the electricity supply to the following equipment when the general body concentration of methane detected in the return air exceeds 2%—

(i) the armoured face conveyor;

(ii) the longwall shearer cutters;

(iii) the longwall equipment in the longwall return airway.

243B Longwall return airway

(1) This section applies in addition to the requirements under section 243.

(2) At least 1 methane detector must be located in each of the following places—

(a) a longwall return airway that is an ERZ0 (a *relevant zone*), if a machine can physically pass through a boundary between—

(i) a NERZ and the ERZ0; or

(ii) an ERZ1 and the ERZ0;

(b) a longwall return airway that is an ERZ1 (also a *relevant zone*), if a machine can physically pass through a

boundary between—

- (i) a NERZ and the ERZ1; or
- (ii) another ERZ1 and the ERZ1.

(3) The detector must automatically activate a visible alarm on the intake side of each access leading into the relevant zone when the general body concentration of methane detected in the return air exceeds 1%.

Note— See section 396.

244 Intake airway intersecting with longwall face

- (1) At least 1 automatic methane detector must be located at the intersection between a longwall face and an intake airway.
- (2) A detector located at the intersection between a longwall face and an intake airway must automatically—
 - (a) activate a visible alarm when the general body concentration of methane detected at the intersection exceeds 1%; and
 - (b) trip the electricity supply to longwall equipment in the longwall face and intake airway when the general body concentration of methane detected at the intersection exceeds 2%.

Note— See section 394.

245 Explosion-protected electrically-powered loader

If a general body concentration of methane exceeding 1.25% is detected around an explosion-protected electrically-powered loader that is not fitted with an automatic methane detector, the loader's operator must switch off the electricity supply to the loader's trailing cable.

246 Explosion-protected vehicle powered by a battery or internal combustion engine

- (1) This section applies to an explosion-protected vehicle—
 - (a) powered by a battery or an internal combustion engine; and
 - (b) fitted with 1 automatic methane detector.
- (2) If a general body concentration of methane of at least 1% is detected around the vehicle, the vehicle's operator must immediately withdraw the vehicle to a place where the general body concentration of methane is less than 1%.
- (3) If a general body concentration of methane of at least 1.25% is detected around a vehicle constructed before 1 July 2001 that is not fitted with a methane detector, the vehicle's operator must immediately switch off the electrical motors or internal combustion engine.

248 Non-explosion-protected vehicle powered by a battery or an internal combustion engine

If the automatic methane detector fitted to a non-explosion-protected vehicle powered by a battery or an internal combustion engine fails in service, the vehicle's operator must immediately park the vehicle.

250 Action to be taken if methane detector activates or is non-operational

(1) An underground mine must have a standard operating procedure for taking action when any of the following happens—

(a) an automatic methane detector fitted to a machine, vehicle or plant to which any of sections 233 to 239 applies—

(i) trips the electricity supply to the machine, vehicle or plant; or

(ii) stops its internal combustion engine;

(b) a methane detector mentioned in paragraph (a), other than a methane detector fitted to a longwall shearer, fails in service;

(c) a methane detector located as required under section 242, 243, 243A, 243B or 244—

(i) fails in service; or

(ii) is being tested or relocated.

(2) The procedure must provide that, if an event mentioned in subsection (1)(a) or (b) happens in relation to a methane detector fitted to a machine or vehicle (other than a longwall shearer or armoured face conveyor) in an ERZ1, the methane detector may be temporarily overridden to allow the machine or vehicle to be moved, but only if—

(a) the general body concentration of methane around the machine or vehicle is less than 1.25%; and

(b) a portable methane detector is used to continuously monitor the concentration.

(3) The procedure must also provide that if an event mentioned in subsection (1)(b) happens in relation to a methane detector fitted to a longwall shearer as mentioned in section 234, the methane detector may be temporarily overridden to allow the longwall shearer to be operated to allow movement to a secure place along the face or at the gate ends, but only if—

(a) the general body concentration of methane around the longwall shearer is less than 1.25%; and

(b) a portable methane detector is used to continuously monitor the concentration.

(4) The procedure must also provide that, if an event mentioned

in subsection (1)(b) happens in relation to a methane detector fitted to a relevant machine or relevant vehicle (other than an armoured face conveyor) being used in a NERZ, the operator may continue to use the machine or vehicle only if—

(a) the general body concentration of methane around the machine or vehicle is less than 0.5%; and

(b) the place where the machine or vehicle is located is continuously monitored by a person using a portable methane detector.

(5) The procedure must also provide that, if an event mentioned in subsection (1)(c) happens in relation to a methane detector located as required under section 243, 243A or 244, the methane detector must be replaced or repaired immediately.

(6) The procedure must also provide that, if an event mentioned in subsection (1)(c) happens in relation to a methane detector located as required under section 242 or 243B—

(a) the methane detector must be replaced or repaired as soon as practicable; and

(b) for a methane detector located in an intake airway at the interface between the zones as mentioned in section 242(1)(a) or (b)—the methane detector may be overridden temporarily to allow operations to continue in the zones until the detector is replaced or repaired, but only if the conditions mentioned in subsection (7) are complied with.

(7) For subsection (6)(b), the conditions are—

(a) a person uses a portable methane detector to continuously monitor for methane—

(i) if the event involves 1 methane detector—at the location of the methane detector; or

(ii) if the event involves more than 1 methane detector at an interface—by moving between the methane detectors at the interface that have failed or are being tested or relocated; and

(b) the electricity supply to the affected zones can be readily tripped when the general body concentration of methane at the location of a methane detector being monitored as required under paragraph (a) exceeds 0.5%.

(8) In this section—

relevant machine means a machine supplied with electricity by a trailing cable.

relevant vehicle means an explosion-protected vehicle powered by a battery or an internal combustion engine.

254 Using aluminium alloys underground

- (1) The underground mine manager for an underground mine must ensure an exposed aluminium alloy is not used underground at the mine if the alloy contains more than 6% by mass of combined magnesium and titanium.
- (2) The underground mine manager must also ensure external rotating or reciprocating parts of plant used underground are not constructed of alloys containing more than 0.6% magnesium and titanium combined.

255 Standard operating procedure

- (1) An underground mine must have a standard operating procedure for the following—
 - (a) storing, transporting, handling and using aluminium alloy objects and equipment underground;
 - (b) disposing of aluminium alloy objects and equipment.
- (2) Without limiting subsection (1), the procedure must have regard to the explosive or combustible nature of accumulated aluminium alloy dust.

256 Belt conveyors

The site senior executive for an underground mine must ensure each belt conveyor used at the mine has the following features—

- (a) an emergency stop system capable of being activated from any point along the length of the conveyor where persons have access;
- (b) certified fire resistant and antistatic conveyor belting and drum lagging.

259 Compressed air equipment

- (1) An underground mine's safety and health management system must provide for managing risk from using compressed air underground.

Example of how the risk may be managed—

by locating compressors on the surface

- (2) The system must provide for the electrical bonding to earth of compressed air equipment, hoses and pipes that are likely, during operation, to develop static electrical charges capable of causing an electric shock to a person or a spark.

Examples of hoses and pipes likely to develop static electrical charges—

1 ventilation ducting

2 a hose that is not antistatic

- (3) This section does not apply to hoses used in the control system of a diesel engine using filtered air.

355 Auxiliary fans

- (1) The ventilation officer for an underground mine must ensure an auxiliary fan used at the mine is located and operated in a way that prevents air recirculating through the fan.
- (2) The ventilation officer must ensure the minimum quantity of air flowing in any panel at the mine is the sum of—
 - (a) the open circuit capacity of each auxiliary fan in operation in the panel; and
 - (b) 30% of the open circuit capacity of the largest auxiliary fan in operation in the panel.
- (3) The site senior executive for the mine must ensure an auxiliary fan, other than a compressed air powered auxiliary fan, used at the mine is capable of switching off automatically if the main ventilation system fails totally.

357 Monitoring fans

- (1) The site senior executive at an underground mine must ensure each fan installed on the surface, and each booster fan installed below ground, at the mine as part of the mine's ventilation system, is fitted with the following—
 - (a) an equipment condition monitoring device;
 - (b) a device capable of continuously monitoring and recording the fan's static pressure.
- (2) The site senior executive must ensure each device mentioned in subsection (1) operates in a way that, if the device detects a significant departure from the fan's normal operating parameters, the device—
 - (a) first, triggers a visible alarm; and
 - (b) second, trips the electricity supply to the fan; and
 - (c) records the date and time of the action mentioned in paragraphs (a) and (b).
- (3) The site senior executive must ensure the part of the device displaying the results of the monitoring is located where the part is easily accessible by a person whose task includes checking the condition of the fan.

370A Activity that is not an on-site activity—[Act, s 10](#)

- (1) For [section 10](#)(2)(g) of the [Act](#), an activity performed by an electricity entity, in relation to its works, and for the purpose of supplying electricity, is not an on-site activity.
- (2) In this section—

electricity entity has the meaning given by the [Electrical Safety Act 2002](#).

works, of an electricity entity, has the meaning given by the [Electrical Safety Act 2002](#), [section 25](#).